Decision Trees

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

Teach tree fitting with complete branch evidence.

Use supplied versus fitted labels explicitly and assess the actual selected cutoff task without claiming it proves fitting mastery.

Keep one habit: Keep development pruning and final evaluation separate; retain missing-input review.

MAIN LESSON · SECTION 1

Prepare the connected experiment

> Keep prior exposure, shared practice and independent evidence distinct.

Goal and main teaching plan

Primary target: Fit a **root** question and **leaf prediction**s from **training** examples

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Materials and prerequisite

Teach two 45-minute sessions. Students fit a **depth**-one decision tree by learning **branch** majorities and selecting the question with the fewest total **training** mistakes.

This transparent classroom algorithm is not a claim to reproduce CART or a library default. Explain tuple positions, list comprehension and the leaves dictionary; offer the full paper route for students who need it.

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

The cutoff was selected using **training** labels. Matching those examples does not establish performance on new examples; validation can expose mistakes and support a baseline comparison.

Printed lesson and workbook

Six **training** row cards and four validation cards

Two **branch** areas labelled 0 and 1

Optional Python 3 editor with no packages required

Opening retrieval

Without opening the old lesson, explain one key idea from Baselines First. Give an example and a mistake that the idea helps you avoid.

Look for: Fit a majority label baseline using **training** labels only

Reference example, not the only acceptable wording: **Training** has four clear and two blocked, so the constant is clear. A tie chooses clear under the declared classroom rule.

Validation labels score the stored **prediction**; using their majority to choose it would change the **fitting** procedure.

Without opening the old lesson, explain one key idea from Leakage. Give an example and a mistake that the idea helps you avoid.

Look for: Choose a **split** unit that matches the intended **prediction** claim

Reference example, not the only acceptable wording: **Training** fits the model; validation compares development choices; test evaluates after choices are fixed.

Repeatedly selecting by the test score uses that set for development, so it is no longer an untouched final evaluation.

Without opening the old lesson, explain one key idea from Inspect Before Modeling. Give an example and a mistake that the idea helps you avoid.

Look for: Inspect row meaning missing values and repeated records

Reference example, not the only acceptable wording: One row is intended to represent one indoor practice trip.

The source is a teacher-authored fictional export for this lesson, duration uses minutes, and None means unknown rather than zero.

End goal

By the end, I can explain decision trees using a traced example and a stated limitation.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Teach the learning cycle

Case exposure check: this source example appears in ai-2/14, ai-2/15, ai-4/11. Reuse a familiar case for recall or modelling; verify that the case used to assess transfer has not already received feedback.

Retrieval answer guidance

Baselines First

Leakage

Inspect Before Modeling

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Rule: if spots>=4 predict spotted, otherwise healthy. Trace 3,4,5 and missing. Does applying this rule demonstrate that **training** found the **threshold**?

Concept to explain

A decision tree partitions feature space using ordered questions; each **leaf** produces a **prediction**.

Term	Meaning
split	question dividing examples
</details>

Term	Meaning
threshold	numeric boundary used in a split
leaf node	terminal prediction
pruning	remove unnecessary branches

Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Readiness answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach root choice, leaf labels and held-out tracing separately

> Avoid inferring an inevitable label from a feature name.

Complete training data and classroom fitting rule

A decision tree asks a feature question and follows a **branch** to a **prediction**. A tree with one question has **depth** one and is sometimes called a stump. Unlike the earlier fixed-threshold examples, where the two **output** labels were fixed in advance, this week we fit both the question and the label stored at each **leaf**.

Our six fictional **training** rows have two binary sensor flags available before **prediction**: near and tall. A value of 1 means that flag is present; 0 means absent. The reviewed scene label is separate.

Each row represents a different trip on a familiar route, including rows with identical values. Similar readings can have different scene labels. These are teaching examples, not robot control instructions.

Row	Near	Tall	Label
1	1	0	blocked
2	1	1	blocked
3	1	0	blocked
4	0	1	clear
5	0	0	clear
6	0	1	blocked

Try near first and tall second. For each question, group the **training** rows by 0 and 1, then predict each group’s majority label; choose clear for tied label counts.

Count all **training** mistakes and keep the first question if scores tie. Stop after one question. This is our stated classroom **fitting** procedure.

Trace scores and algorithm limits

Feature 0 is near; feature 1 is tall; row[2] is the label. **Output** is 0 1, then 1 2, then 0 clear blocked. Near makes one **training** mistake: near=0 contains two clear labels and one blocked, so that **leaf** predicts clear.

Near=1 contains three blocked labels. Tall makes two mistakes because both its groups have a blocked majority.

The stored tree contains the chosen feature and its two learned leaves. The code assumes nonempty data, binary flags, allowed labels and examples on both **branch**es of each candidate.

An empty **branch** would need a declared fallback; this small example does not implement one. The score is total **training** mistakes, not an average giving small and large **branch**es equal weight.

This is an explicitly limited **depth**-one learner. General tree algorithms can repeat **split** selection within **branch**es and often use impurity criteria such as Gini. Our code does not implement that full procedure.

Keeping the scope small lets you inspect every decision.

Term	Meaning
Root	The first question asked by a tree
Branch	A path selected by the answer to a question
Leaf	An endpoint that stores a prediction
Depth	The largest number of question steps from root to leaf

Trace fixed validation predictions and baseline

Two paths through the learned tree

ROOT QUESTION	BRANCH ANSWER	LEARNED LEAF
Is near equal to 1	No near is 0	Predict clear
Is near equal to 1	Yes near is 1	Predict blocked

Both paths start at the same root. Tall is not used by this fitted tree; the leaf labels came from training counts.

Validation rows are (near=1,tall=1,blocked), (0,0,clear), (0,1,blocked), (1,0,clear). Follow only near: **prediction**s are blocked, clear, clear, blocked. Two of four match.

The **training**-majority baseline predicts blocked, since **training** has four blocked and two clear; it also matches two of four validation labels.

> **Which validation rows does the tree miss, and which does the baseline miss? Explain why the tall value cannot change this tree's prediction.**

Keep the fitted **root** and leaves unchanged while scoring. Equal accuracy does not mean identical errors, and five correct **training prediction**s out of six is not the validation result.

Read the separate spot-rule picture

Trace a small fitted tree, including its exact boundary.

Toy stored tree: if visible_spots>=2 predict spotted; otherwise healthy. Missing visible_spots requires review. **Input**s L=1, M=2, N=None.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

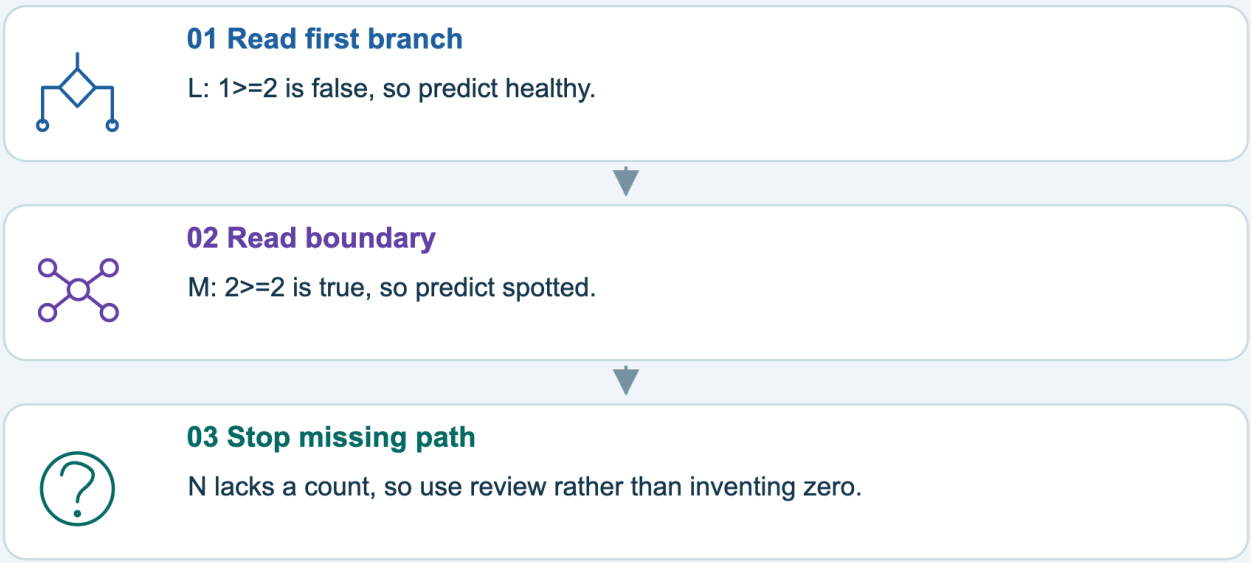
Main fitted tree chooses near and stores near0→clear,near1→blocked. On its four validation rows the tree missesrows3,4, while the blocked baseline missesrows2,4: both2/4but different errors. Separate picture uses a supplied visible_spots>=2 rule with review for missing; the exact count2distinguishes>=2from>2. Neither picture nor boundary tracing proves that2was fitted or optimal.

The independent assessment uses its own cutoff5.

Trace exact boundary and changed operator

AI ACADEMY / AI-4/11 · 02 · TRACE THE CASE

Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Read first branch	L: 1>=2 is false, so predict healthy.
2. Read boundary	M: 2>=2 is true, so predict spotted.
3. Stop missing path	N lacks a count, so use review rather than inventing zero.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Read the picture in order. N lacks a count, so use review rather than inventing zero. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Applying a stored tree is inference. Selecting a **threshold** from labelled **training** data is a separate **fitting** step.

Change: The stored rule changes to visible_spots>2.

Prediction: M is now predicted healthy.

Reason: Changing >= to > moves the exact value 2 to the other **branch**.

▼ Additional supplied cutoff3example

A supplied tree predicts spotted for spot_count>=3, otherwise healthy.

Trace 2,3,4 and state boundary.

Healthy, spotted, spotted. Three is included. This supplied rule does not show an optimal learned **threshold**.

▼ Earlier wording, alternatives and picture transcripts

Explain the mechanism

Trace the information, fitted behavior, and evaluation proof separately.

QUESTION Trace, draw, and prune a tiny tree while preserving test behavior.

MODEL IDEA A decision tree partitions feature space using ordered questions; each **leaf** produces a **prediction**.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD **Root** asks whether spot_count ≥ 3. Yes **branch** predicts spotted; no **branch** asks whether yellow_ratio ≥ 0.25.

Draw a fresh data → model → **prediction** → evaluation trace:

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

STEP 1 **Root** asks whether spot_count ≥ 3.

STEP 2 Yes **branch** predicts spotted; no **branch** asks whether yellow_ratio ≥ 0.25.

STEP 3 Trace **leaf** A: spot_count=1, yellow_ratio=0.10 → healthy.

Use development validation evidence when comparing pruning choices; once the final test guides changes, it is development evidence and cannot remain an untouched final check.

CHECK Verify the denominator, unit, **split**, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

Concrete teaching case

Exact explanation to model

Explain the added worked example

Key terms

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

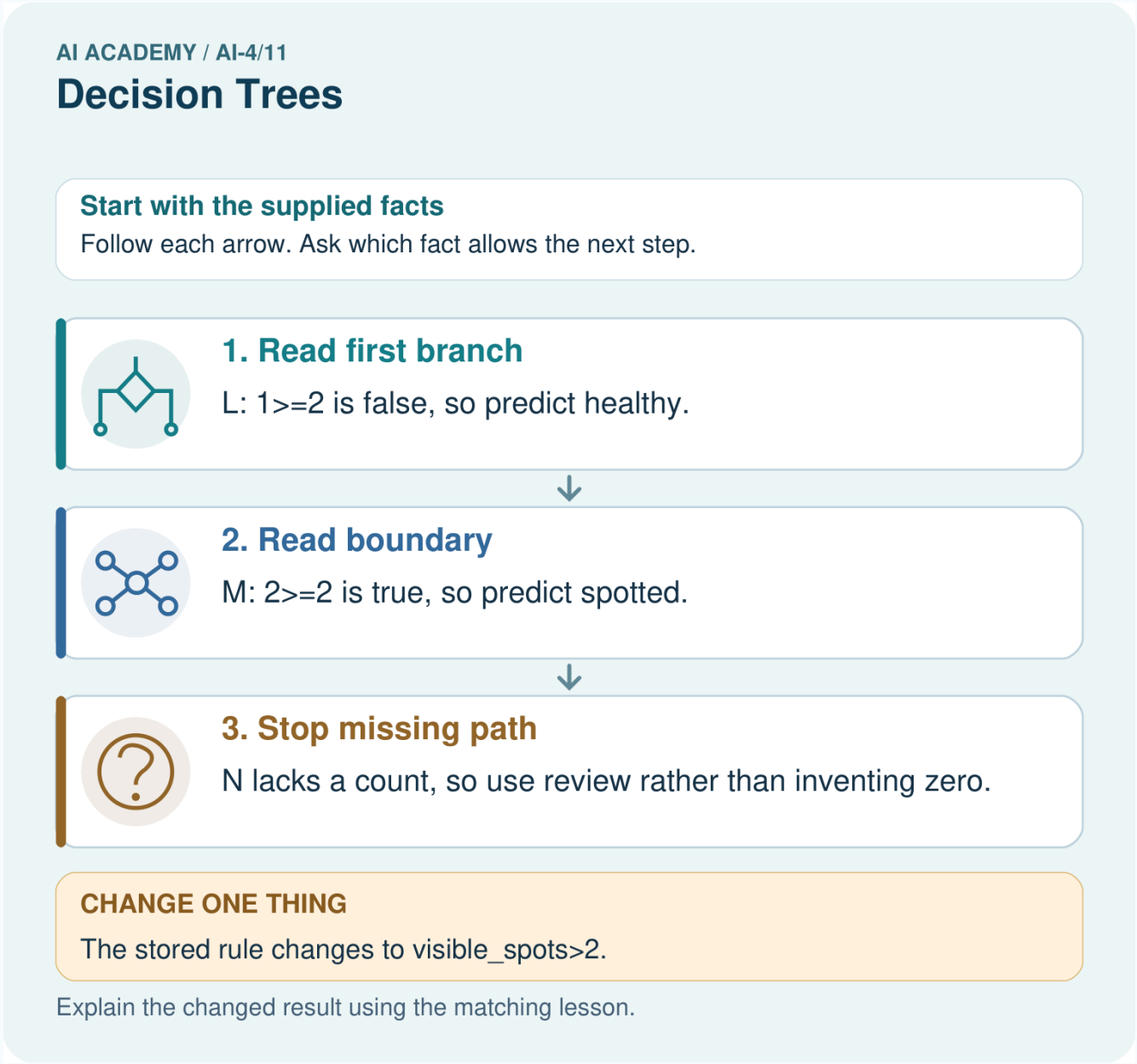
These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Decision Trees

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Guide the calculation, then remove help

> Collect each learner's prediction before showing its key.

Current completion task and checked key

Completion task: For each near **branch** and each tall **branch**, count blocked and clear labels and choose a **leaf**. Give the total errors for each candidate question.

Teacher check: Near=0 has 1 blocked and 2 clear, predicts clear with 1 error; near=1 has 3 blocked and 0 clear, predicts blocked with 0 errors. Tall=0 has 2 blocked and 1 clear; tall=1 also has 2 blocked and 1 clear.

Both tall leaves predict blocked with 1 error each. Totals are near 1 and tall 2.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for **Fitting** a decision tree. Proposed explanation: “A near **branch** must predict blocked”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: The **branch** label is learned from its **training** labels; W4 deliberately reverses that relationship.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial question and changed-case reasoning

Which **input** distinguishes the original and changed tree? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

M with exactly 2 spots. Values 1 and missing keep their earlier declared paths; 2 is the boundary probe. Changed-case check: M is now predicted healthy. Changing >= to > moves the exact value 2 to the other **branch**.

Changed-case reasoning: M is now predicted healthy. Changing >= to > moves the exact value 2 to the other **branch**.

Collect each learner’s explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Keep each learner explaining

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

▼ Earlier wording, alternatives and picture transcripts

Compact alternative: this two-session sequence covers selected core tasks. Use the three-session weekly plan for the full lesson, independent transfer, feedback and project work; allow extra time for extensions.

Demonstrate and guide practice

Board sequence

Write the given facts: A supplied tree predicts spotted for spot_count>=3, otherwise healthy.

Write the question to explain: Trace 2,3,4 and state boundary.

Reveal after **prediction**s: Healthy, spotted, spotted. Three is included. This supplied rule does not show an optimal learned **threshold**.

Use the earlier transfer question as additional practice. Reserve the fresh stored-cutoff 5 boundary and missingness task for the first individual attempt and record support used.

Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

Model these guided questions

Guided: tree **prediction**s blocked,clear,clear,blocked miss rows 3 and 4. Baseline blocked misses rows 2 and 4. Both get 2/4 or 50 percent. Tall cannot affect **prediction** because the stored **root** uses near only.

Exit: the near=1 **leaf** contains three blocked **training** examples, but the whole tree gets 2/4 on this tiny validation set and misses one near=1 validation case. This does not establish reliability on future trips.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve training search versus validation. Introduce root, branch, leaf and depth with one question.

Stage	Teacher action
7 to 18 minutes	Sort the six cards by near. Count labels within each branch and assign its majority without using validation.
18 to 32 minutes	Repeat for tall and complete W1 and W2. Add branch mistakes to score the full training set.
32 to 45 minutes	Trace code and the diagram. Predict all four validation rows, complete the guided question and compare the baseline.
Session 2 start to 8 minutes	Retrieve why a leaf label is learned rather than dictated by the feature name.
8 to 20 minutes	Complete W3 and discuss depth limits, tied questions and the assumptions required by the short code.
20 to 35 minutes	Collect W4 independently before hints. Require branch counts, both candidate scores, learned leaves and validation comparison.
35 to 45 minutes	Use W5 and the exit claim. Give the fresh retry after feedback and record support.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Keep predictions, observed or supplied outcomes and unmeasured cases separate.

Supplied inputs and checked guidance

Core fitting: near makes 1 error; tall makes 2. Training correctness is 5/6; validation is 2/4. The baseline also gets 2/4 with different errors. Missing flags need review under the declared complete-input scope. Optional full/pruned trees score 3/4 versus 4/4 on the separate development fixture, not a final test. Selected independent answers: 4 healthy; 5 and 6 spotted; missing review.

Inference is not fitting, and final-test labels cannot select the cutoff. The previous near/tall refit W4 remains additional practice.

Predict, investigate and record

1. 1. Write the hypothesis and expected **evidence** before running.
2. 2. Human Tree Builder: choose **split**s, trace six examples, compare train/test errors, then prune one weak **branch**.
3. 3. Run one normal, one boundary, one missing, and one shifted case.
4. 4. Calculate the metric from raw counts or row-level errors.
5. 5. Interpret what changed, what did not, and what remains unsafe or unknown.

Keep the experiment ledger

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose the current misconceptions

A near **branch** must predict blocked: The **branch** label is learned from its **training** labels; W4 deliberately reverses that relationship.

Every feature is used during **prediction**: Only questions in the stored tree are followed. Candidate questions rejected during **fitting** do not affect this tree's **output**.

A deeper tree must work better: More questions can fit **training** detail. Future performance needs separate evaluation; **depth** choices must be documented.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Keep the test conditions fixed, record every result, and change only one planned factor.

Make the lab runnable

Use the supplied complete investigation for shared practice. Reserve the fresh stored-cutoff 5 boundary and missingness task for the individual first attempt before hints or feedback; earlier transfer questions remain additional practice.

Diagnose and reteach

MISCONCEPTION TO REPAIR A deeper tree must be better because it asks more detailed questions.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the selected experiment transfer

> Keep this key until after a first attempt; record any help and prior exposure.

Selected independent task and checked key

Selected case: A stored fitted rule says spots >=5 predicts spotted; otherwise healthy. Trace4,5,6 and a missing count. Policy: missing count requires review. Can test labels be used to choose this cutoff?

4 predicts healthy;5 and6 predict spotted; missing requires review. Applying the stored cutoff is inference, not **fitting**. Choosing it using test labels contaminates the final evaluation; use designated development data.

Judge trace, scope and support

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Name the parts and decisions

Identify the **root**, its two **branch**es, **leaf prediction**s and **depth** of the worked fitted tree. Which parts were selected using **training** labels?

The **root** asks near, with **branch**es 0 and 1. Leaves predict clear for 0 and blocked for 1. **Depth** is one. **Training** labels determined both **leaf** majorities and the error comparison that selected near rather than tall.

W2 Verify the candidate scores

For each near **branch** and each tall **branch**, count blocked and clear labels and choose a **leaf**. Give the total errors for each candidate question.

Near=0 has 1 blocked and 2 clear, predicts clear with 1 error; near=1 has 3 blocked and 0 clear, predicts blocked with 0 errors. Tall=0 has 2 blocked and 1 clear; tall=1 also has 2 blocked and 1 clear.

Both tall leaves predict blocked with 1 error each. Totals are near 1 and tall 2.

W3 Separate fitting from prediction

A learner sees a blocked validation case with near=0 and changes that **leaf** to blocked before reporting the score. Explain what changed and how to report the original fitted tree fairly.

The **leaf** was changed using validation **evidence**, so it is no longer the original **training**-fitted tree. Score the original stored leaves on all the same validation rows and record any later development change separately.

Preserve a separate final test for final evaluation.

W5 Audit a tie and a claim

If near and tall make the same **training** error count, which question does this code retain and why? Does retaining that question prove it will be better on future trips?

It retains near because near is tried first and strict < replaces only for fewer errors. This is a declared tie convention, not **evidence** of better future performance. Separate evaluation and relevant examples are still needed.

▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

Worked case answer

Healthy, spotted, spotted. Three is included. This supplied rule does not show an optimal learned **threshold**.

Additional transfer practice question

A tree predicts fern when **leaf**_width_cm <= 3 and grass otherwise. Trace widths 2, 3 and 4. Which case checks the boundary?

Reasoned practice answer

Widths 2 and 3 predict fern; 4 predicts grass. Width 3 checks the inclusive boundary.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification **threshold**.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “An inclusive **threshold** excludes an example exactly equal to it.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Outputs are healthy, spotted, spotted, and review or an explicitly defined missing-value path. Tracing applies a rule.

To claim **training**, show how candidate **threshold**s were evaluated on **training** examples and how the chosen **threshold** was selected.

Explanations for the additional workbook examples

W1 Name the parts and decisions

W2 Verify the candidate scores

W3 Separate fitting from prediction

W4 Fit a new tree independently

New **training** rows (near,tall,label) are (1,0,clear),(1,1,clear),(1,0,clear),(0,1,blocked), (0,0,blocked),(0,1,clear). Use the same candidate order and tie policies. Give both candidates' leaves and errors, then select the tree.

Validation rows are (1,1,clear),(0,0,blocked),(0,1,clear),(1,0,blocked). List tree **prediction**s and compare accuracy with the **training**-majority baseline.

Near=0 predicts blocked from 2 blocked and 1 clear; near=1 predicts clear from 3 clear. Near has 1 error. Both tall groups have 2 clear and 1 blocked, so both leaves predict clear and total errors are 2.

Choose near with reversed **leaf** labels compared with the worked example. Validation **prediction**s clear,blocked,blocked,clear match rows 1,2: 2/4. Baseline clear matches rows 1,3: 2/4. Both are 50 percent but errors differ.

W5 Audit a tie and a claim

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, support and revisit the experiment

> Retain failures and assess a fresh parallel case after help.

Project evidence

Draw the current Plant Classifier tree and annotate feature, **threshold**, **prediction**, and test **evidence** at each **leaf**.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Support and selected reassessment

Use a tiny labelled dataset, explicit units and a named positive class. Stretch: require a baseline, held-out evaluation and subgroup or shifted-condition analysis with denominators.

Use physical sorting before code. Explain row[feature] as selecting a flag and row[2] as reading its label. Supply empty **branch**-count tables.

Extension: explain why an empty **branch** needs a fallback, and why this code would silently return clear if given an empty label list.

Selected independent task is stored spots>=5, not refitting the older near/tall W4:4→healthy,5/6→spotted,missing→review. Applying it is inference; choosing cutoff from final-test labels contaminates evaluation. This alone does not establish fitting mastery. For a separate uncoached fitting retry, **training**(0,0,clear),(0,1,blocked),(1,0,clear),(1,1,blocked) givesnear2errors(clearleaf ties),tall0errors(tall0clear,tall1blocked). Require both tie policies and a fresh validation trace before claiming fitting transfer.

Delayed return

Suggested return points in this level: ai-4/12, ai-4/14, ai-4/17. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge and reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 12: Nearest Neighbors. Retrieve today's **evidence** and identify what carries forward and what is new.

Next is Week12: Nearest Neighbors. Carry forward the **training**/validation **split** and baseline. Primary reference: https://scikit-learn.org/stable/modules/generated/sklearn.tree.DecisionTreeClassifier.html ; its library criteria differ from this classroom total-mistake fitting procedure.

▼ Earlier wording, alternatives and picture transcripts

Connect and extend learning

Responsible Plant Classifier

Support and stretch

Exit reflection

Bridge to the next week

Support and fresh reassessment

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route without repeating the whole cycle.

Planning and assessment reference

Teaching target: Fit a **root** question and **leaf prediction**s from **training** examples

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

Before class, solve the supported practice: For each near **branch** and each tall **branch**, count blocked and clear labels and choose a **leaf**. Give the total errors for each candidate question.

Reasoned practice answer: Near=0 has 1 blocked and 2 clear, predicts clear with 1 error; near=1 has 3 blocked and 0 clear, predicts blocked with 0 errors.

Tall=0 has 2 blocked and 1 clear; tall=1 also has 2 blocked and 1 clear. Both tall leaves predict blocked with 1 error each. Totals are near 1 and tall 2.

Check every learner before discussion: Use the worked case for **Fitting** a decision tree. Proposed explanation: “A near **branch** must predict blocked”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: The **branch** label is learned from its **training** labels; W4 deliberately reverses that relationship.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: The stored rule changes to visible_spots>2. Require a **prediction** and a reason before comparing. Expected result: M is now predicted healthy. Reason: Changing >= to > moves the exact value 2 to the other **branch**.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback and evidence records**

REFERENCE · SECTION 8

Reference · Feedback and evidence records

> Keep supported rehearsal separate from independent performance.

Record and repair

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

▼ **Earlier wording, alternatives and picture transcripts**

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The **branch**-boundary and indexing lessons deliberately include failure cases.

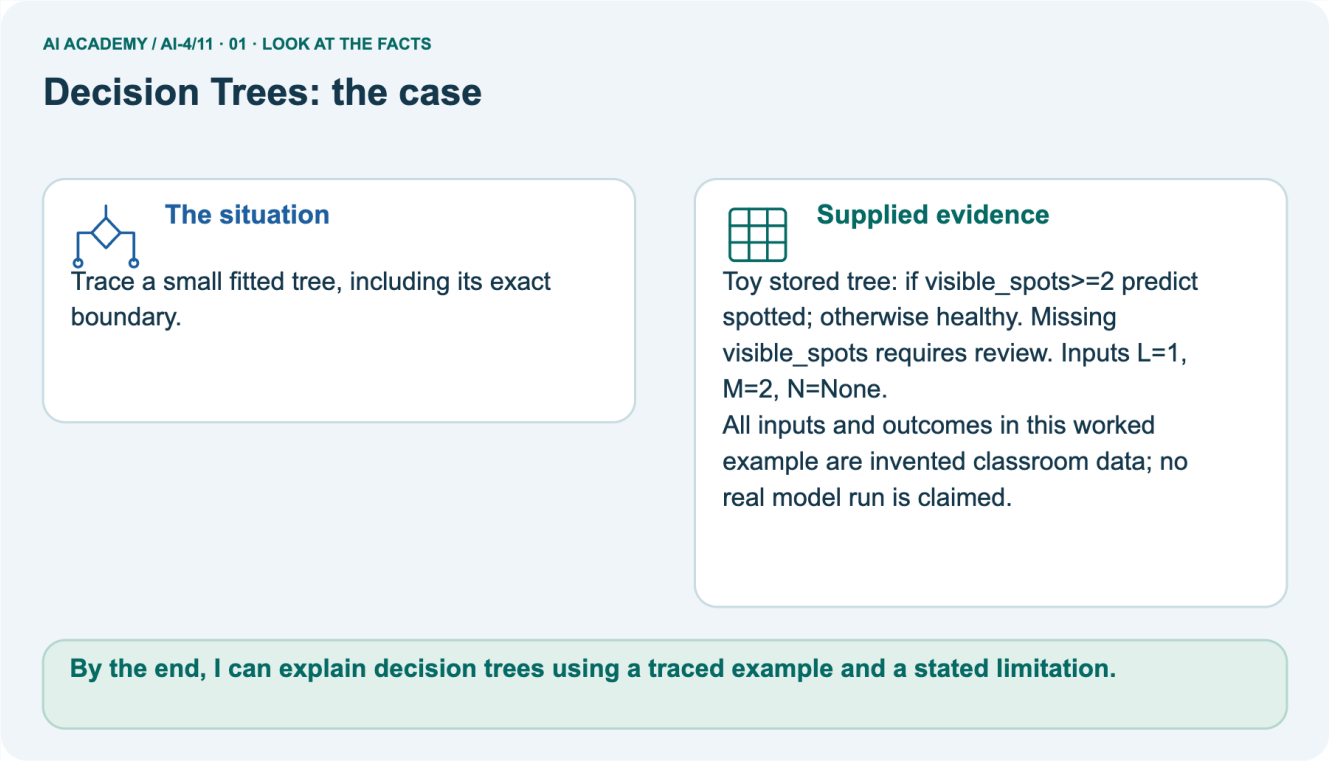
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a specific reasoning move.

01. Decision Trees: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

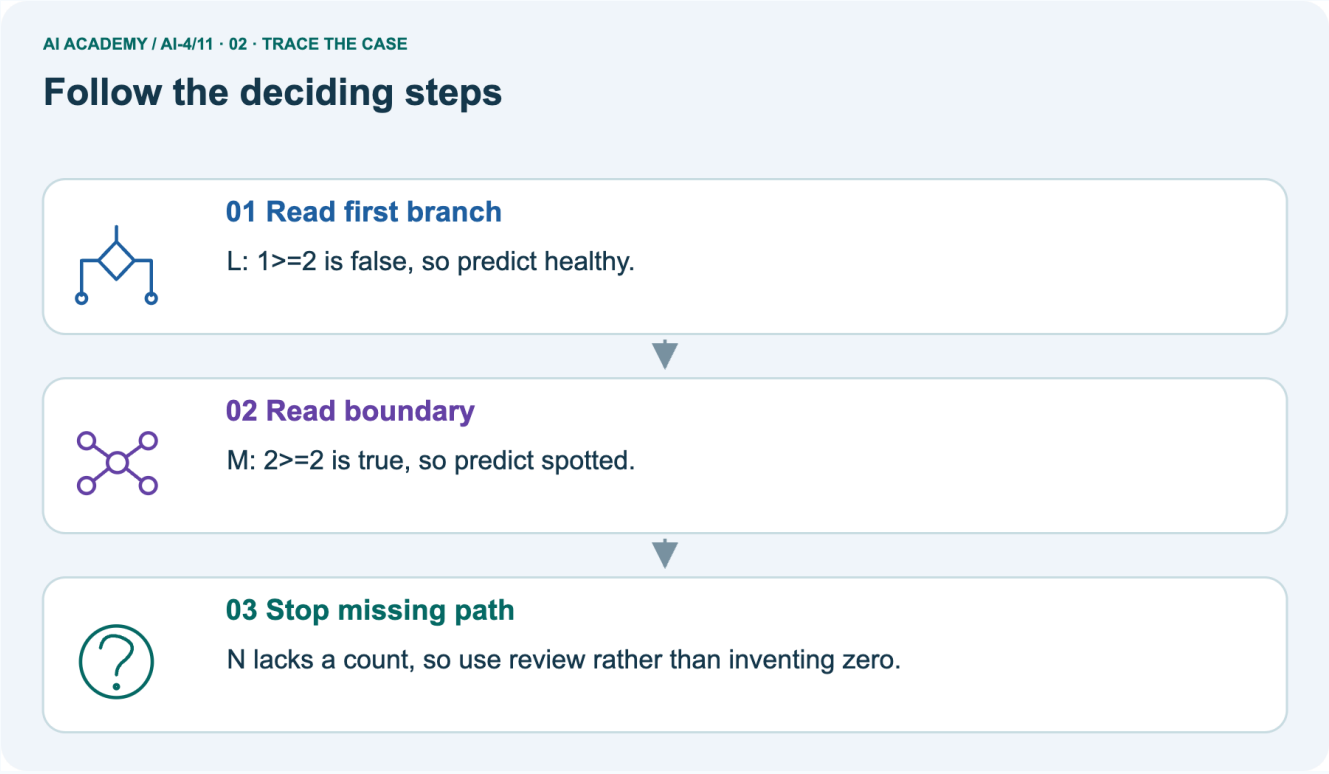
Trace a small fitted tree, including its exact boundary. Supplied facts: Toy stored tree: if visible_spots>=2 predict spotted; otherwise healthy. Missing visible_spots requires review. **Input**s L=1, M=2, N=None.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



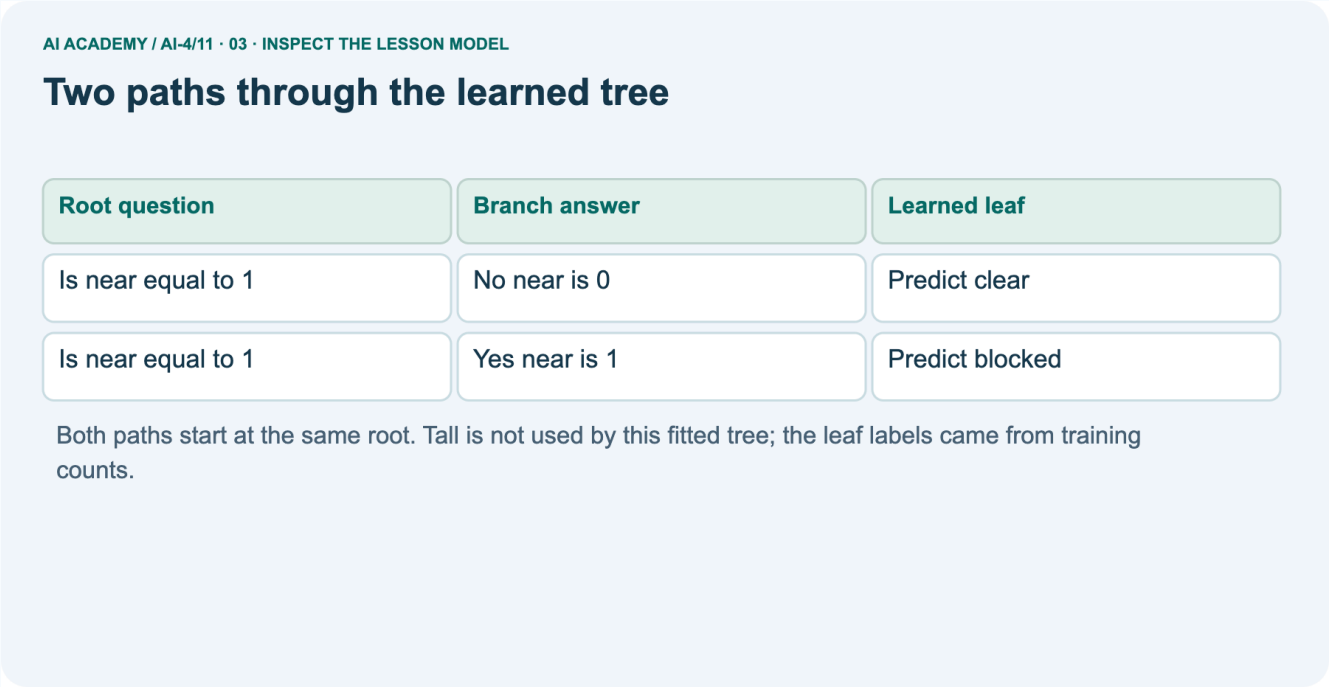
Follow the labelled picture with the matching explanation. Open the vector diagram

Read first **branch**: L: 1>=2 is false, so predict healthy. → Read boundary: M: 2>=2 is true, so predict spotted. → Stop missing path: N lacks a count, so use review rather than inventing zero.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Two paths through the learned tree



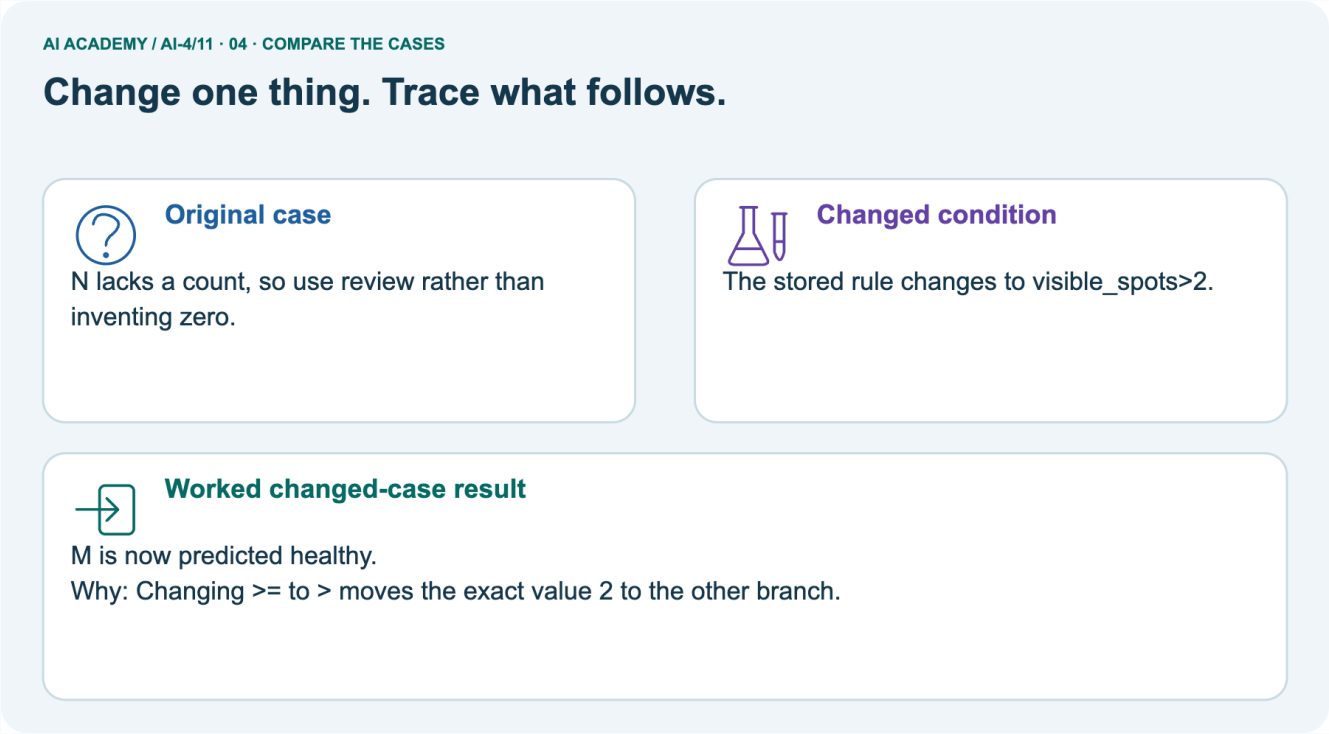
Follow the labelled picture with the matching explanation. Open the vector diagram

Two paths through the learned tree. Is near equal to 1 | No near is 0 | Predict clear. Is near equal to 1 | Yes near is 1 | Predict blocked

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: N lacks a count, so use review rather than inventing zero. Changed condition: The stored rule changes to visible_spots>2. Expected result: M is now predicted healthy.

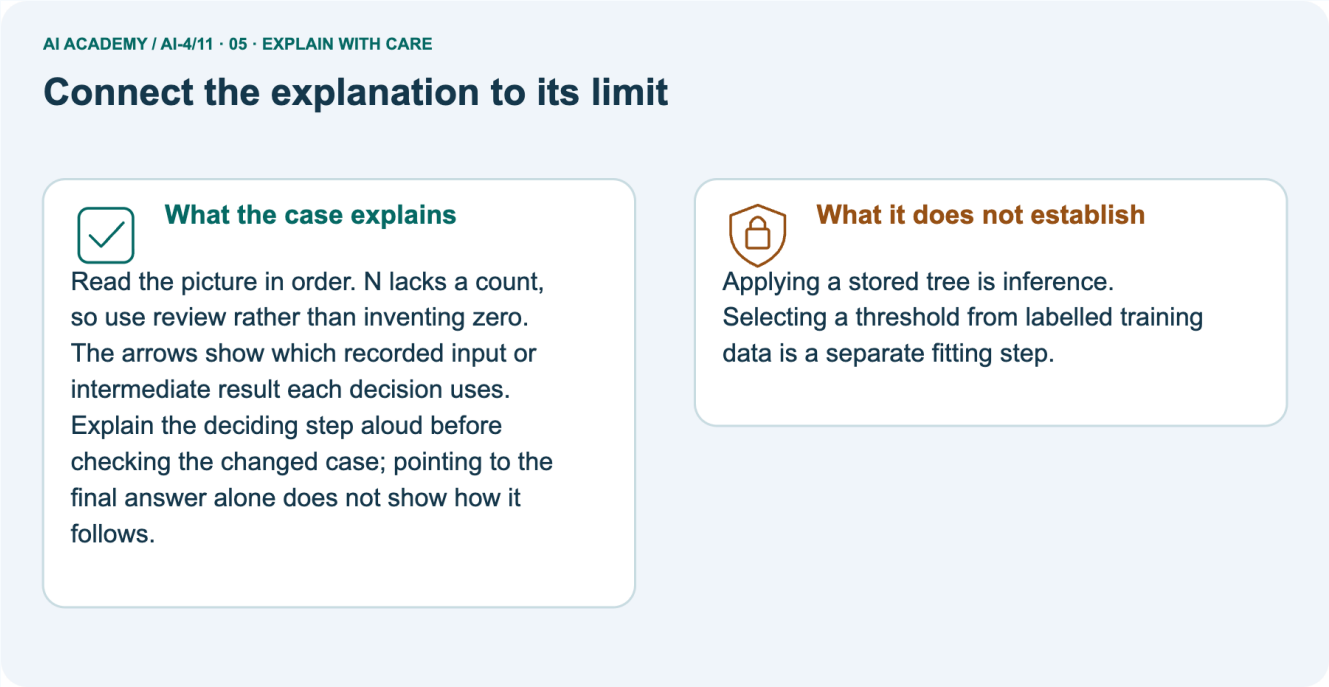
Reason: Changing >= to > moves the exact value 2 to the other **branch**.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Changing >= to > moves the exact value 2 to the other **branch**.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. N lacks a count, so use review rather than inventing zero. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows. Limit: Applying a stored tree is inference.

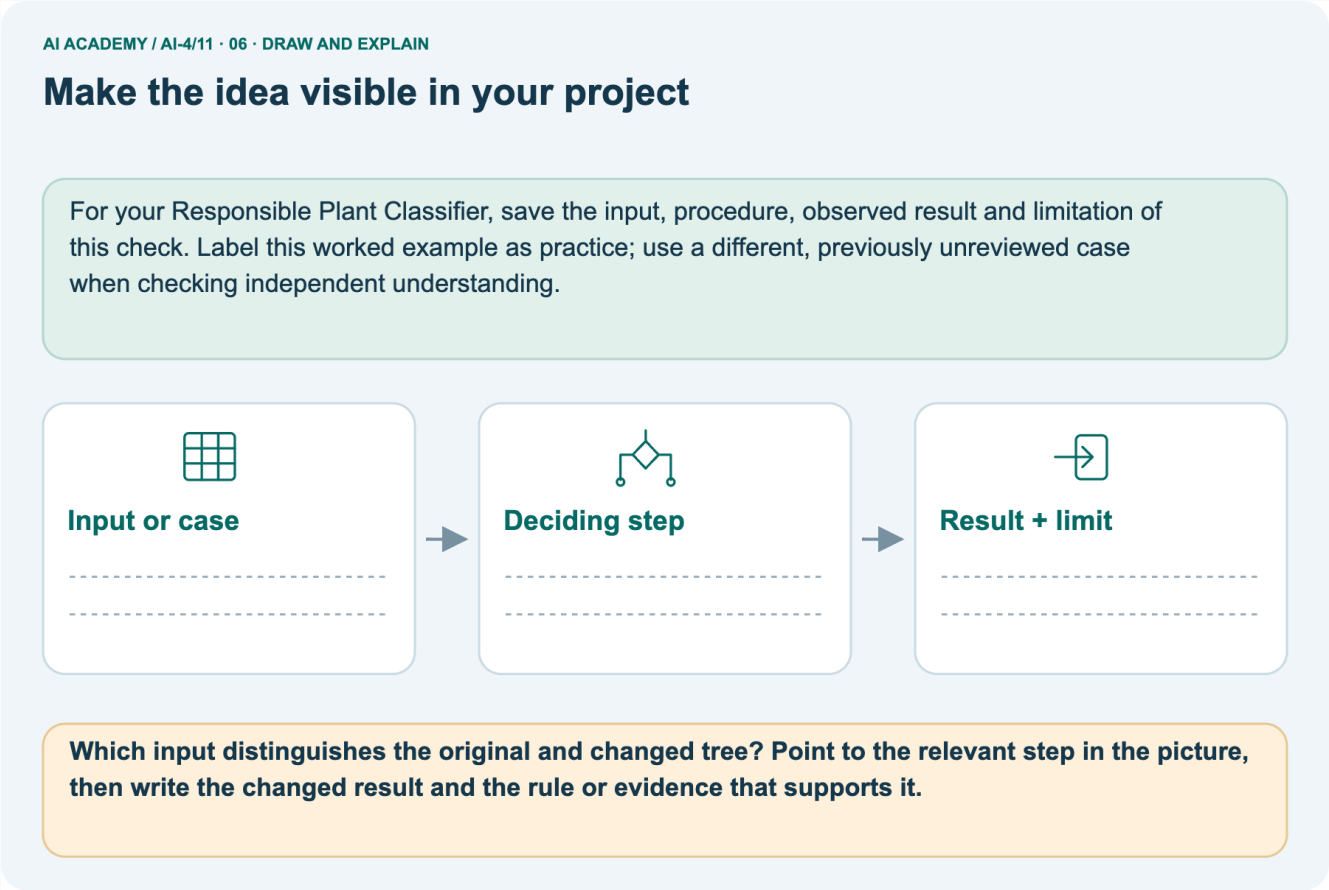
Selecting a **threshold** from labelled **training** data is a separate **fitting** step.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Applying a stored tree is inference. Selecting a **threshold** from labelled **training** data is a separate **fitting** step.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which **input** distinguishes the original and changed tree? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: M with exactly 2 spots. Values 1 and missing keep their earlier declared paths; 2 is the boundary probe. Changed-case check: M is now predicted healthy.

Changing >= to > moves the exact value 2 to the other **branch**.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Decision Trees: the case

02. Follow the deciding steps

03. Two paths through the learned tree

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.